

FINAL PROJECT (30 Marks)

I. PROJECT OVERVIEW:

Students will choose an organization in Karachi (e.g., retail, healthcare, finance, education, etc.) and work through the entire process of data warehousing and data mining. They will collect data, design a data warehouse model, apply data mining techniques, and visualize the results.

II. STEPS FOR THE PROJECT:

Step	Details
1. Selecting an Organization	- Choose a Karachi-based organization willing to provide operational data (e.g., sales, customer details, employee performance). Examples: • Retail outlets (sales data) • Healthcare facilities (patient records) • Banks (transactional data) • Educational institutions (student performance)
2. Data Collection	- Collaborate with the organization to gather relevant business data. May include: • Historical sales data • Customer demographics and behavior • Employee workload/performance data • Other operational datasets
3. Data Preprocessing and ETL	- Clean and preprocess the data: • Handle missing/inconsistent values • Convert formats (e.g., categorical to numerical) • Apply ETL: Extract data → Transform → Load into data warehouse
4. Data Warehousing Design	- Design the data warehouse schema: • Star Schema: Facts and dimensions • Snowflake Schema: Normalized dimensional structure • Fact Tables: Quantitative data (e.g., revenue, transactions) • Dimension Tables: Descriptive data (e.g., customers, products, time)
5. Data Mining	- Apply suitable techniques:

Techniques	• Classification: Predict categories (e.g., churn) • Clustering: Group similar records • Association Rules: Identify co-purchased items • Regression: Predict numerical trends (e.g., future sales)
6. Data Visualization	- Visualize findings using tools: • Bar Charts: Compare categories (e.g., sales/product) • Pie Charts: Show proportions (e.g., customer segments) • Line Charts: Trends over time • Heatmaps: Correlations • Scatter Plots: Relationships (e.g., budget vs revenue)
7. Reporting	- Compile a complete project report with: • Introduction: Organization overview and problem • Data Collection: Sources and methods • ETL: Preprocessing description • Data Warehouse: Schema and tables • Mining: Techniques and findings • Visualizations: Charts with interpretation • Conclusion: Key insights and business recommendations

III. SUBMISSION GUIDELINES:

Section	Details
1. Formatting	- Font: Times New Roman or Arial - Font Size: 12 pt (body), 14 pt (headings) - Line Spacing: 1.5 - Margins: 1-inch on all sides - Page Numbers: Footer (centered) - Headings: Use Heading 1, Heading 2 styles consistently
2. Data Gathering Evidence	- Source Documentation: • Letter from the organization • Screenshots/records of data sources (Excel, DB, etc.) • Signed consent forms if applicable - Data Samples: Include sample data in the appendix - Privacy: Anonymize data; remove PII
3. Plagiarism Policy	- Original Work: Must be student's own - Citation Style: APA format - Plagiarism Check: Report will be checked via software - Avoiding Plagiarism: Paraphrase with credit, use quotes with proper citation

4. Project Submission	- Deadline: May 15th, 2026 - Format: • Report: Submit as a Word file • Data Files: Submit as a zipped folder (.zip) • Visualizations: Embedded in report or as separate files (PNG/JPEG) - Naming Conventions: • Report: DW_DM_Project_Report_YourName.pdf • Data: DW_DM_Project_Data_YourName.zip • Visuals: DW_DM_Project_Visualizations_YourName.zip
5. Presentation	- Slides: Prepare PowerPoint summarizing methodology, analysis, findings - Duration: 5–10 minutes - Submission: Upload separately along with report and data

IV. FINAL SUBMISSION FORMAT OF THE DOCUMENT

Section	Content Details
1. Title Page	- Project Title - Student Name(s) - Course Name and Code - Instructor Name - Submission Date
2. Abstract	- A concise summary of the entire project (150-200 words). - Briefly mention the organization, data gathering process, techniques used, and key findings.
3. Introduction	- Project Overview: Briefly describe the organization chosen and the business problem they are addressing. - Objective of the Project: State the goal of the project (e.g., applying data warehousing and mining techniques to solve a business problem).
4. Data Collection	- Data Sources: Description of the data sources obtained from the organization (e.g., sales data, customer data, transactional data). - Data Gathering Process: Explain how the data was collected. - Evidence of Data: Include sample data or screenshots from the organization.
5. ETL Process	- Extract: Describe how data was extracted from its source. - Transform: Discuss the transformation processes (e.g., cleaning, normalization, encoding).

	- Load: Explain how the data was loaded into the data warehouse.
6. Data Warehouse Design	- Schema Design: Explain the schema used (star or snowflake schema), along with the fact tables and dimension tables. - Schema Diagram: Include a visual representation of the schema (ER diagram).
7. Data Mining Techniques	- Techniques Used: Describe the data mining techniques applied (e.g., classification, clustering, regression, association rules). - Tools Used: Mention any software/tools used (e.g., Python, Tableau, Power BI). - Analysis: Provide a summary of the findings obtained through data mining.
8. Data Visualization	- Visual Representations: Include charts, graphs, heatmaps, or other visualizations to represent your data findings. - Interpretation: Explain what each visualization represents and how it ties into the business problem.
9. Results & Analysis	- Key Findings: Summarize the main insights derived from the data mining process. - Business Implications: Discuss how these findings can be used by the organization for decision-making.
10. Conclusion	- Summary of Insights: Provide a brief overview of the key findings. - Recommendations: Suggest any recommendations for the organization based on the results of the data analysis.
11. References	- List all sources cited in the report using APA citation style. - Include books, articles, websites, and any data sources used.
12. Appendices	- Appendix A: Raw and cleaned data files (if permissible by the organization). - Appendix B: Additional visualizations not included in the main report. - Appendix C: Any additional supporting documentation, such as consent forms, data source confirmations, etc.